QUANTITATIVE METHODS 1

Worksheet 7 – Integrated cases

Exercises

Case A – Continuous-assessment preparation

A school surveys 200 students about the maximum price they would accept for an optional service:

Price	$[5, 10[$	$[10, 15[$	$[15, 20[$	$[20, 30[$	$[30, 40[$	$[40, 60[$
n_i	10	25	45	60	40	20

Identify population, unit and variable type; build the complete table with widths, densities, frequencies, ICC, DCC and midpoints; draw histogram and cumulative curves; calculate modal class, median, quartiles, mean, population variance, standard deviation and CV; find the price affordable to about 65%; then classify a satisfied/not satisfied variable and choose a graph.

Case B – Final-exam preparation

Part 1 – Regression

$$x = (4, 5, 7, 8, 10, 11, 13, 15), \qquad y = (8, 10, 14, 15, 20, 22, 26, 31).$$

Calculate r, $\hat{y} = ax + b$, R^2, residuals and the forecast at $x = 18$. State whether it is interpolation or extrapolation and explain why correlation does not prove causality.

Part 2 – Time series

For $80, 100, 120, 140, 88, 110, 132, 154$, calculate 4-quarter moving averages and centred moving averages, describe the overall pattern and explain centring.

Part 3 – Indices

Using products X and Y from Worksheet 6, calculate I_P^F, I_Q^F, I_V, verify factor reversal and deflate 2.4 million euros when the price index is 120.